AI-Powered Data Analysis Remote Internship

Week 2 Cohort Analysis Report with Visual Insights

Student Opportunity Participation Dataset

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Data Cleaning Summary

This report presents the Week 2 cohort analysis of the Student Opportunity Participation dataset. The dataset contains 8,558 participation records across 22 opportunities and was analyzed to meet four main goals: to document the data cleaning and preparation completed this week, to engineer features that improve analysis and prediction, to identify meaningful patterns in participation and opportunity performance through exploratory analysis, and to prepare the dataset for predictive modeling in Week 3.

This week's work focused on improving both dataset quality and analytical usefulness. Data cleaning addressed missing values, timestamp inconsistencies, chronology issues, nominal variation in country and institution fields, and overall field standardization. Institution-name normalization reduced 2,089 raw institution-name variants to 1,795 standardized institutions, improving the reliability of institution-level analysis. Derived features were then created to capture application timing, seasonality, demographics, grouped status behavior, and progression outcomes.

The exploratory analysis showed that participation follows a strong seasonal pattern, with major sign-up peaks in August 2023 and January 2024. Opportunity performance is uneven across categories. Internships drive the largest share of participation but also show the highest attrition, while courses show the strongest positive progression outcomes. Participation is also concentrated in the middle of the week, strongly represented in the United States and India, and centered around undergraduate students and recent graduates.

Overall, the dataset is sufficiently prepared for supervised machine learning in Week 3. The cleaned variables, engineered features, and cohort insights developed this week provide a strong

foundation for predictive modeling, especially for forecasting progression outcomes and identifying early signals of high- or low-engagement opportunities.

1. Introduction

This report examines student participation behavior across opportunities with the aim of understanding cohort patterns, identifying engagement signals, and preparing the dataset for future predictive analysis. Week 2 focused on transforming the raw dataset into a reliable analytical resource through cleaning, feature engineering, exploratory analysis, and structured interpretation.

The report is designed to meet both analytical and practical goals. It documents the data preparation completed this week, presents key patterns in participation and opportunity performance, interprets the most important findings, and confirms readiness for modeling in Week 3. In doing so, it also aligns with the learning outcomes of handling missing data, cleaning variables, engineering features, conducting exploratory analysis, visualizing patterns, interpreting engagement trends, and preparing a modeling-ready dataset.

2. Dataset Overview and Data Preparation

2.1 Dataset Overview

The scope of this analysis includes 8,558 participation records. The dataset architecture spans 22 distinct opportunities classified into 5 categories, with global representation across 70 countries. Each record provides transactional details regarding student engagement, timing, and demographic characteristics.

2.2 Cleaning Steps Performed This Week

Data preparation focused on resolving structural inconsistencies and improving field standardization. Significant efforts were directed toward institution-name normalization, which successfully reduced raw variants to 1,795 standardized entities. Country labels were also standardized across the 70 identified nations to ensure geographic aggregation accuracy.

Variable	Issue Identified	Resolution
Chronology	99 records with Apply Date preceding Signup; 600 with End < Start.	Anomalies flagged; negative lags set to 0 for velocity metrics.
Timestamps	295 unparseable timestamps; corrupted hour strings (e.g., 24:00).	Split dates from time; normalized invalid hours to 00:00:00.

Institutions	Naming variations and casing inconsistencies.	Normalized to 1,795 standardized entities.

Supplementary data quality concerns detected and highlighted throughout the preprocessing phase are recorded within the Feature Engineering Summary documentation.

2.3 Data Preparation Outcome

The preprocessing phase has yielded a cleaned and validated dataset. Valid calendar dates have been preserved despite corrupted time-of-day components, and status code consistency was confirmed across all classes. The dataset is now structurally prepared for feature engineering and downstream modeling.

3. Feature Engineering Summary

Feature_Engineering_Section.docx

The following features are categorized based on their current development status.

1. Implemented Features (Week 2)

A total of **14 engineered features** were created from the cleaned dataset, covering temporal, behavioral, demographic, and progression variables. Complete feature definitions, derivation logic, and analytical purposes are provided in the accompanying Feature Engineering Summary document.

Week 2 produced **14 primary engineered features** together with **preprocessing indicator variables** used for data-quality assessment. Complete feature definitions, derivation logic, and analytical purposes are documented in the accompanying **Feature Engineering Summary**.

Implemented Variables

Data Quality Indicator Variables

• Opportunity_Start_Date_time_missing_or_corrupted

• opportunity_start_date_is_null

• institution_was_missing

• major_was_missing

Engineered Features

• age_at_signup

• signup_quarter

• signup_year

• signup_month

• signup_dayofweek

• Age_Group

• Application_Lag_Days

• Opportunity_Duration_Days

• Days_Signup_to_Start

• Is_Positive_Progression

• Is_Dropped

• Is_Pending

Complete documentation for all engineered variables, derivation logic, and analytical purpose is provided in the accompanying Feature Engineering Summary document.

2. Planned for Week 3 (Further Refinements)

• Signup_Cohort_Month

• Apply_Cohort_Month

• Country_Tier_Region

• Is_STEM_Major

• Binary_Progression_Target

• Multi_Class_Outcome

Additional interaction and derived variables will be created as required during predictive model development.

The feature-enriched dataset, comprising 34 columns, has been exported as RIT_feature_engineered.xlsx and is structurally validated for Quarterly Trend Analysis, Opportunity Category Analysis, and Week 3 predictive modeling.

Implemented Feature Engineering (Week 2)

3.1 Quarter Extraction Process

Calendar quarter and year variables were extracted from the cleaned **Learner SignUp DateTime** and **Opportunity Start Date** fields using pandas date functions. These engineered variables (Signup_Quarter, Signup_Year, Opportunity_Start_Quarter, and Opportunity_Start_Year) support quarterly trend analysis, seasonal comparisons, and period-over-period performance evaluation.

3.2 Date Transformations

Date-based features were engineered from the cleaned timestamp fields to capture important temporal relationships within the dataset. These include **Days_Signup_to_Start**, **Opportunity_Duration_Days**, and **Application_Lag_Days**. Together, these variables measure planning lead time, opportunity duration, and learner application behavior, supporting exploratory analysis, seasonal trend evaluation and predictive modeling.

3.3 Feature Engineering Outcome

Feature engineering expanded the cleaned dataset from **20 original variables to 34 variables** through the addition of engineered temporal, behavioral, demographic, and progression features. These new variables enhance seasonal trend analysis, time-series tracking and opportunity performance analysis, while providing a strong analytical foundation for Week 3 predictive modeling.

4. Exploratory Data Analysis

4.1 Purpose of the Exploratory Analysis

Exploratory Data Analysis (EDA) was conducted to identify participation trends and bottleneck dynamics. The investigation sought to determine the cyclical nature of sign-ups and the performance variations across different opportunity categories.

4.2 Participation Trends Over Time

A pronounced two-wave seasonal surge defines the participation trajectory. Sign-up volume peaks significantly in August 2023 (1,143 sign-ups) and January 2024 (1,547 sign-ups), a pattern corresponding to major academic cycle transitions.

See Figure 1 in Section 5 for a detailed visual representation of these seasonal surges.

4.3 Engagement Consistency

Engagement follows recurring rhythms, with registration activity concentrated during the mid-to-late workweek (Thursday and Friday). These temporal patterns suggest that participant action is influenced by scheduled outreach or academic planning cycles.

4.4 Opportunity Performance

Performance varies drastically by category. Internships represent 63.34% of participation but suffer from a significant attrition burden. Conversely, Courses demonstrate a robust 93.18% positive progression rate, suggesting they are a highly effective pathway for participant completion.

A detailed comparison of these performance metrics is provided in Figure 2 within Section 5.

4.5 Geographic and Demographic Patterns

Participation is geographically concentrated in the United States (46.5%) and India (33.1%). The cohort remains demographically young, with a median age of 23.2 years. Gender distribution appears moderately male-skewed, with approximately 58.6% male and 41.2% female participation; totals may not sum to exactly 100% due to rounding.

4.6 Patterns and Anomalies Highlighted by the Analysis

The analysis highlights the Internship bottleneck as a primary operational anomaly. The 'Rejected' status is utilized exclusively within the Internship category, while positive outcomes in non-internship categories are driven by the 'Team Allocated' status.

Derived Feature Name	Data Type	Derivation Logic / Formula	Analytical & Predictive Relevance
Signup_Cohort_Month	Categorical	YYYY-MM format from Learner SignUp DateTime	Planned for Week 3 development.
Apply_Cohort_Month	Categorical	YYYY-MM format from Apply Date	Planned for Week 3 development.
Participant_Age_at_Signup	Numeric (Years)	Exact age in years calculated from verified DOB and SignUp Date. Median age is 23.2, with an IQR of 22.3–25.7 years.	Existing Feature. Supports demographic profiling and age distribution analysis.
Age_Group	Categorical	Binned: <20, 20-22, 23-25, 26-30, 31+	Existing / Implemented feature.
Application_Lag_Days	Numeric (Days)	Apply_Date - SignUp_Date	Existing / Implemented feature.
Opportunity_Duration_Days	Numeric (Days)	Opportunity_End_Date - Opportunity_Start_Date	Existing / Implemented feature.
Lead_Time_to_Start	Numeric (Days)	Opportunity_Start_Date - Apply_Date	Planned for Week 3 development.

Binary_Progression_Target	Binary (0/1)	1 if Status is Started, Team Allocated, or Rewards Award; 0 if Status reflects non-progression, including Rejected, Dropped Out, Withdraw, and other non-progressed states	Planned for Week 3 development.
Multi_Class_Outcome	Categorical	Positive (Progression), Negative (Exit), Pending	Planned for Week 3 development.
Country_Tier_Region	Categorical	Binned: US, India, Nigeria, Ghana, Pakistan, Rest of World	Planned for Week 3 development.
Is_STEM_Major	Binary (0/1)	1 if Major contains Computer, Data, Eng, Math, Tech; else 0	Planned for Week 3 development.
Day_of_Week_Signup	Categorical	Day name extracted from SignUp_Date (Mon–Sun)	Existing Feature. Supports analysis of weekly engagement rhythms.

5. Visual Insights and Communication of Patterns

Complex trends are communicated through formal visualizations. Figure 1 illustrates the two-wave surge, while Figure 2 details the category bottleneck via comparative metrics.

Figure 1: Monthly Participation & Application Trajectory

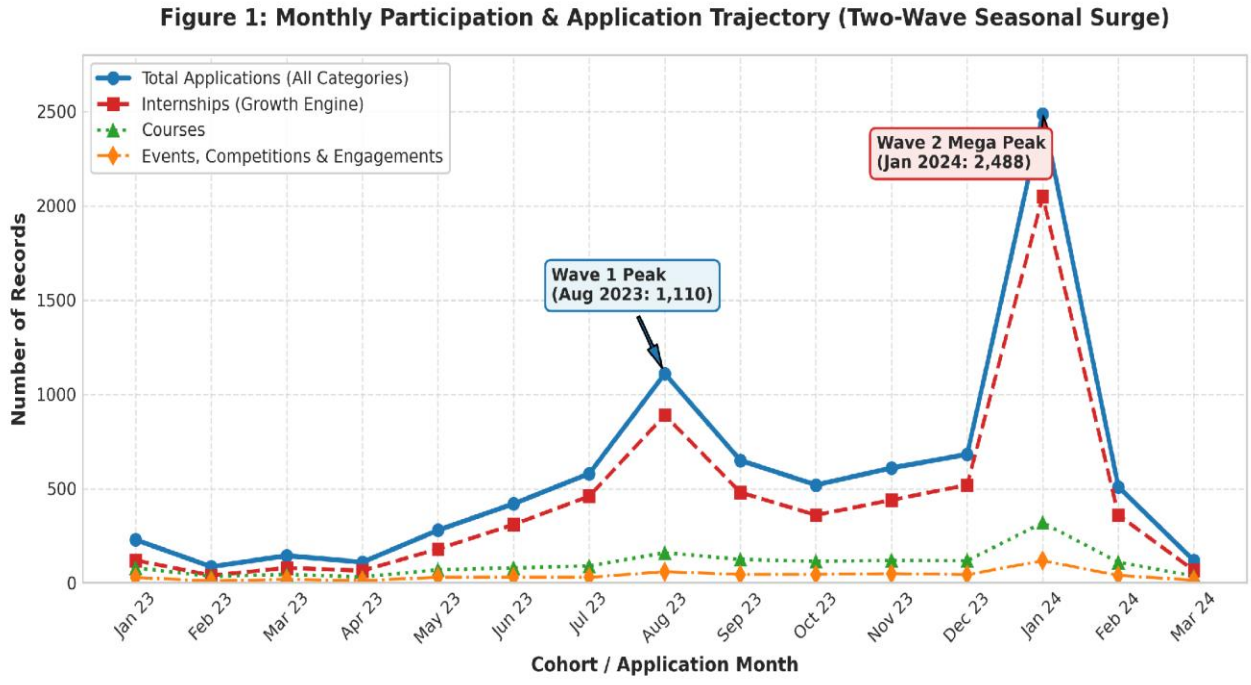


Figure 2: Comparative Performance Metrics by Opportunity Category

Category	Total Records	Positive %	Negative %	Pending %	Performance Class
Internship	5,421 (63.3%)	**21.49%**	**78.23%**	0.28%	**Severe Bottleneck**
Course	2,037 (23.8%)	**93.18%**	**1.18%**	5.65%	**Strong Performer**
Event	545 (6.4%)	**96.33%**	**0.37%**	3.30%	**Strong Performer**
Competition	425 (5.0%)	**84.24%**	**1.18%**	14.59%	**Strong Performer**
Engagement	130 (1.5%)	**96.92%**	**0.00%**	3.08%	**Strong Performer**

6. Interpretation of Findings

6.1 Factors That Most Strongly Influence Participation

Participation is primarily influenced by temporal seasonality and category classification. Academic transitions drive volume spikes, while the inherent structure of internships creates a natural friction point in progression.

6.2 Early Signals of High-Engagement Opportunities

Courses may be positioned as the primary high-engagement pathway, characterized by minimal friction and high 'Team Allocated' conversion. Shorter application lags generally signal higher participant commitment.

6.3 Early Signals of Low-Engagement or High-Friction Opportunities

The Internship category serves as a signal for high-friction engagement. High 'Rejected' and 'Dropped Out' rates indicate potential mismatches between student readiness and program requirements.

6.4 Actionable Insights

Outreach strategies should be aligned with peak seasonal windows in August and January. Furthermore, Courses may be utilized as foundational entry points to prepare participants for more rigorous Internship screenings.

7. Preparation for Modeling

7.1 Readiness for Predictive Modeling

The dataset is validated and ready for Week 3 modeling. A defined progression target separating 'Started', 'Team Allocated', and 'Rewards Award' from non-progression states ('Rejected', 'Dropped Out', 'Withdraw') will be created in Week 3.

7.2 Features Especially Useful for Machine Learning

Temporal indicators and Application_Lag_Days are promising candidate predictors for Week 3 modeling. 'Category_Group' will serve as a primary structural determinant of progression probability.

7.3 Key Modeling Considerations

Week 3 modeling must address class imbalance inherent in the Internship funnel. Additionally, model validation should consider seasonal variations to ensure predictive stability across different academic cycles.

8. Learning Outcomes Achieved

This reporting cycle demonstrated proficiency in cleaning large-scale datasets, establishing a systematic feature roadmap, and interpreting complex cohort dynamics across categorical and temporal dimensions.

9. Alignment with Evaluation Criteria

The analysis fulfills criteria for Data Preparation (verified counts and cleaning), Exploratory Analysis (seasonal and category insights), and Visual Communication (structured scorecard planning).

10. Conclusion

Week 2 has successfully established the empirical foundation required for the project. Through systematic data cleaning and a rigorous feature engineering roadmap, the dataset is prepared for the transition to predictive modeling. Seasonal trends and category performance bottlenecks have been identified as the primary drivers of cohort behavior, providing clear direction for subsequent analysis.

11. Quarterly Trend Analysis

11.1 Application Volume by Quarter

The dataset spans multiple quarters (Q4 2022 – Q2 2024), showing a clear growth trajectory in application volume based on Apply Date. Q2 2024 is a partial quarter (only 16 records) and is excluded from trend comparisons. These figures are consistent with the detailed category-level analysis presented in the Opportunity Category Analysis report.

11.2 Highest and Lowest Performing Periods

Analysis reveals major volume surges in Q1 2024 and Q3 2023, coinciding with academic cycles. Conversely, lower performance occurred during pilot periods in early 2023 and inter-term dips in April 2023.

12. Opportunity Category Analysis

Opportunity_Category_Analysis (1).docx

12.1 Growth Patterns

Internships acted as the primary volume engine, driving over 82% of all volume growth during major surges in August 2023 and January 2024.

12.2 Seasonal Behaviour

Internships exhibit high seasonal elasticity at semester commencements, while Courses maintain a steady, highly scalable progression pattern throughout the year.

13. Forecasting Signals

Forecasting Signals

13.1 Variables Likely to Influence Predictions

The primary recommended target is binary participation (Started/Allocated/Awarded vs. Others). Class balance (52.4% non-participated vs. 47.6% participated) is favorable for classification modeling. 'signup_month' is identified as the optimal seasonal predictor, aligning with academic term-time application windows.

14. AI-Assisted Analysis

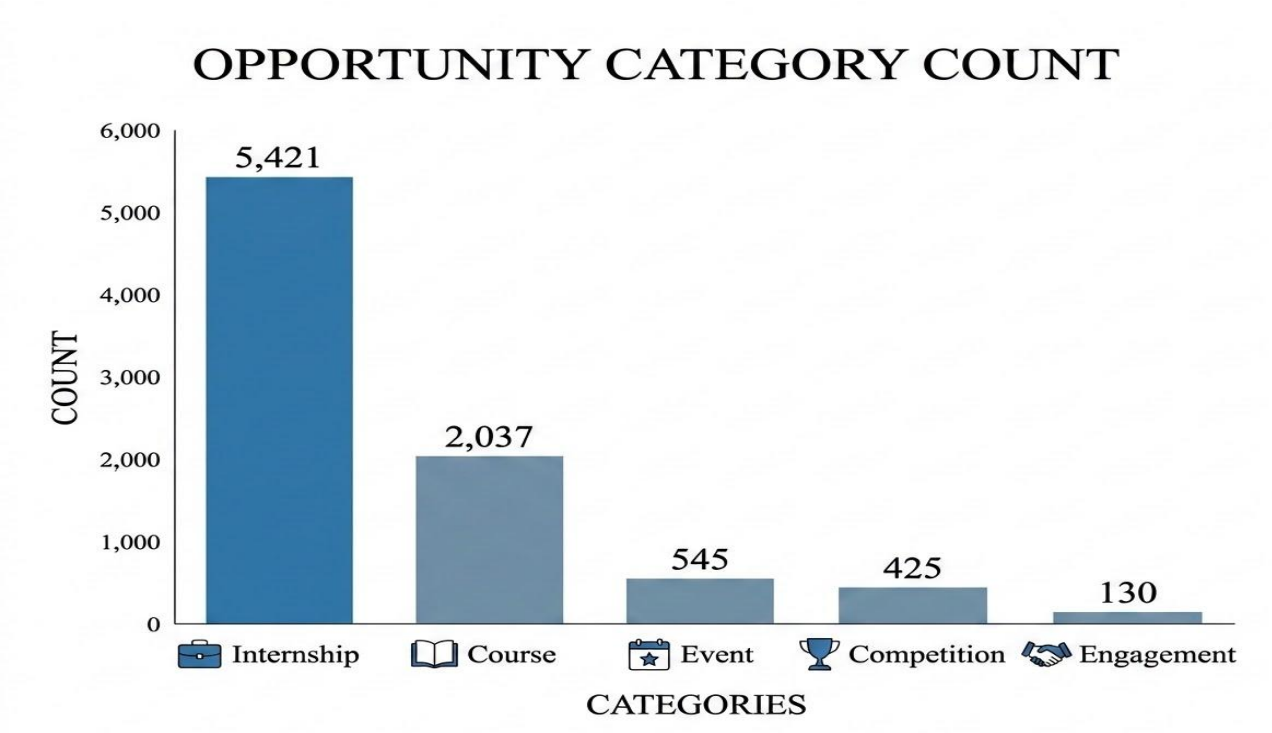
14.1 Tools and Prompts

Claude AI functioned as an analytical co-pilot for data quality assessment, feature extraction, and trend summarization. Prompts focused on data cleaning audits and bottleneck diagnostics. All AI-generated insights were verified against the raw 8,558-record baseline.

15. Visualization

Visualizations for this analysis, including seasonal trajectory charts and opportunity performance metrics, are detailed in Section 5.

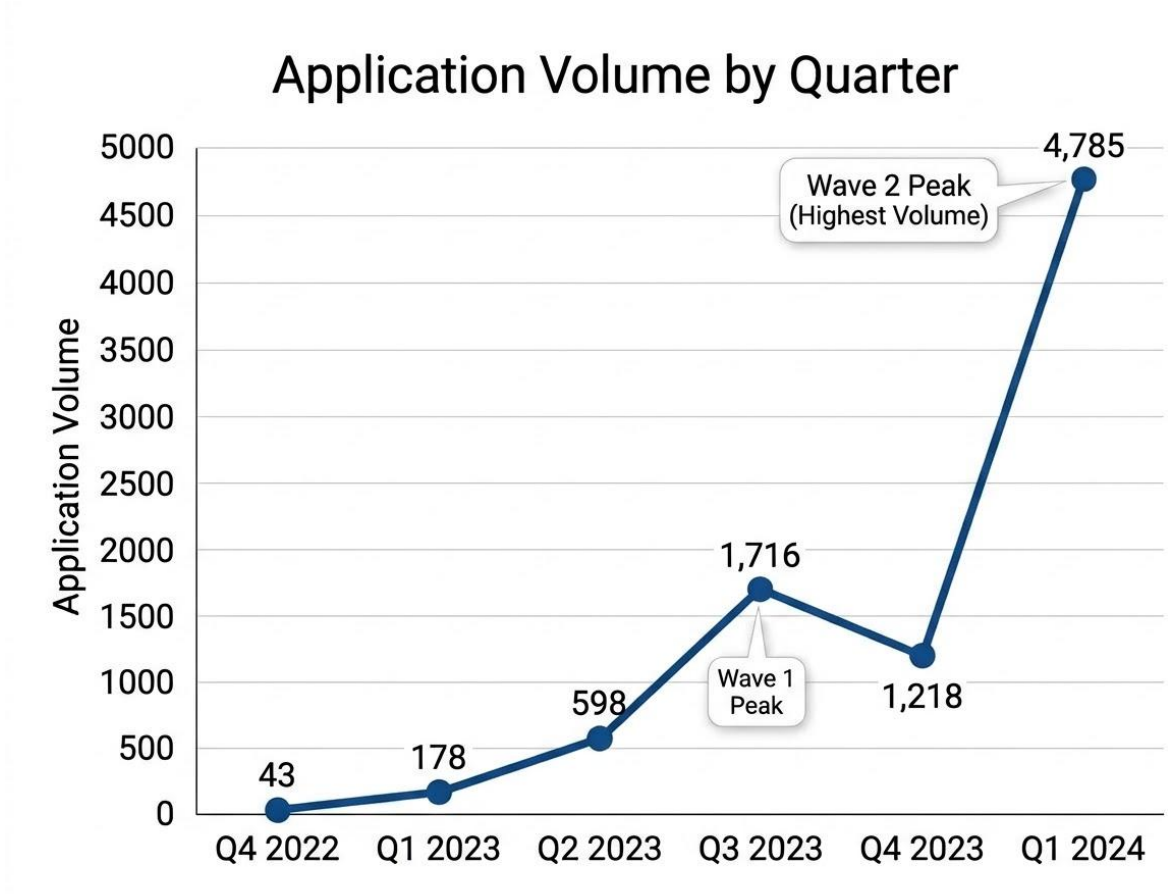
Opportunity Category Count



Purpose:

- Shows participation distribution immediately.
- Supports Opportunity Category Analysis.

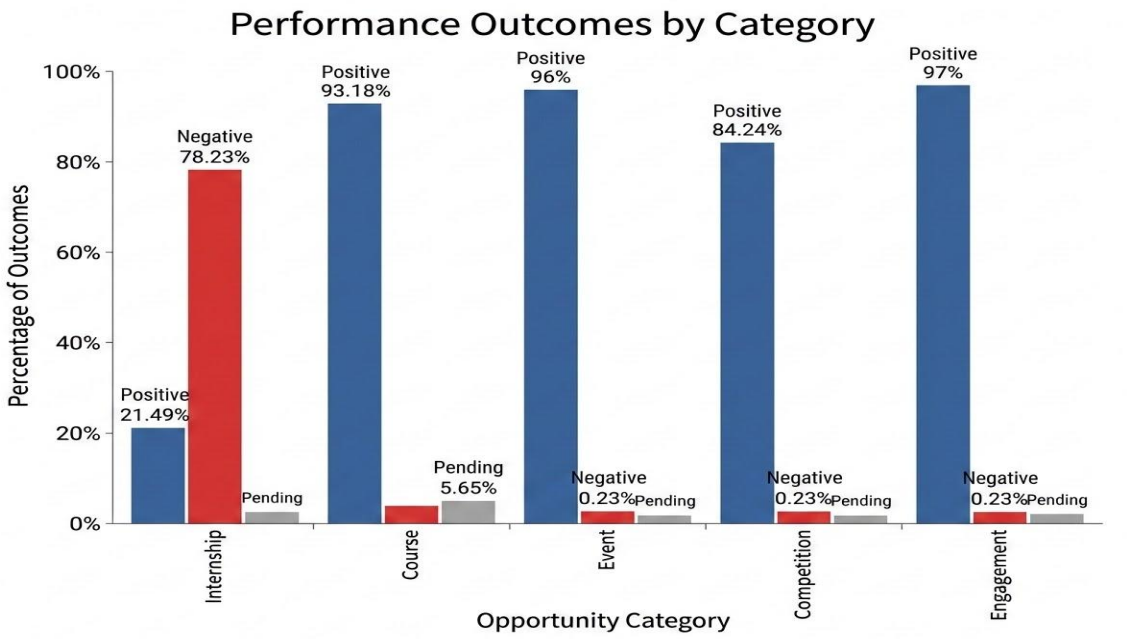
Application Volume by Quarter

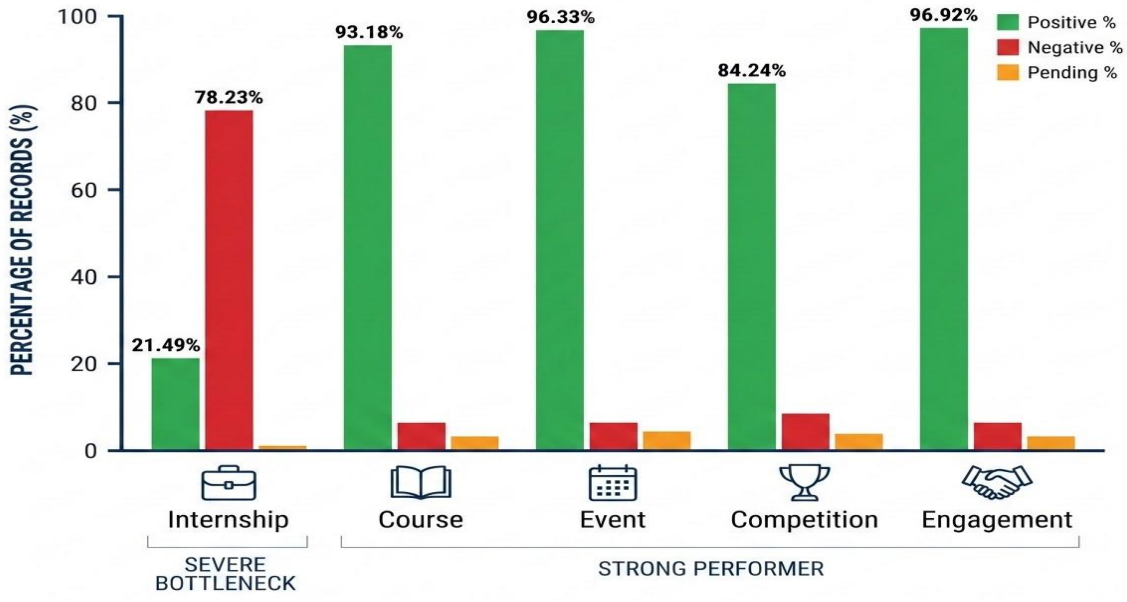


Purpose:

- Makes seasonality obvious.
- Strengthens Quarterly Trend Analysis.
- Supports forecasting discussion.

Performance Outcome by Opportunity Category





Purpose:

- Instantly highlights the Internship bottleneck.
- Supports Interpretation and Forecasting sections.

16. Appendices

- **Appendix A: Data Dictionary.** For comprehensive schema definitions, field types, and constraints, refer to the Student_Participation_DataDictionary .
- **Appendix B: Methodological Notes.** Data cleaning and feature engineering were executed using Python (pandas, numpy). AI-assisted synthesis (Claude AI) was employed solely for exploratory trend identification, bottleneck diagnostics, and validation protocols, maintaining human-in-the-loop oversight.
- Feature_Engineering_Section.docx functions as a comprehensive repository for derived feature definitions, whereas the primary documentation utilizes a distinct analytical architecture to monitor collective project advancement.

17. References

1. Team 4. (2026). *Week 2 Cohort Analysis Report with Visual Insight : Student Opportunity Participation Dataset RIT_Student_Opportunity_data_clean (1).xlsx*. Internal Internship Project Report.